

1. compose.yml

```
1  version: "3.8"
2
3  services:
4    nginx:
5      image: "phoeng/nginx:1.1.0"
6      container_name: api-gateway
7      ports:
8        - "8000:8000"
9      networks:
10       - todo-net
11      depends_on:
12        - todo-service
13        - notification-service
14    todo-service:
15      image: "todo:release-2"
16      container_name: todo-service
17      environment:
18        REDIS_HOST: redis
19        NOTIFICATION_HOST: notification
20      networks:
21        - todo-net
22
23    notification-service:
24      image: "todo-notification:release-2"
25      container_name: notification-service
26      environment:
27        REDIS_HOST: redis
28      networks:
29        - todo-net
30
31    webapp:
32      image: "todo-webapp:release-1"
33      container_name: webapp
34      stdin_open: true
35      ports:
36        - 3000:3000
37      networks:
38        - todo-net
39    redis:
40      image: "redis"
41      container_name: redis
42      networks:
43        - todo-net
44
45  networks:
46    todo-net:
47      driver: bridge
48      # internal: true
49
```

2. Docker compose up -d

```
(base) PS C:\Users\USER\Desktop\Software_def\act3\todo-compose> docker compose up -d
time="2026-08-25T20:14:48+07:00" level=warning msg="C:\\Users\\USER\\Desktop\\Software_def\\act3\\todo-compose\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 6/6
✔ Network todo-compose_todo-net Created 0.1s
✔ Container notification-service Started 1.3s
✔ Container redis Started 1.4s
✔ Container webapp Started 1.3s
✔ Container todo-service Started 1.1s
✔ Container api-gateway Started 1.6s
(base) PS C:\Users\USER\Desktop\Software_def\act3\todo-compose>
```

3. Docker ps -a

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
9477b37da95	phoeng/nginx:1.1.0	"/docker-entrypoint..."	About a minute ago	Up About a minute	0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
000/tcp	api-gateway				
5ed95cb24478	redis	"docker-entrypoint.s..."	About a minute ago	Up About a minute	6379/tcp
	redis				
45ea2a0c4e31	todo:release-2	"/bin/sh -c 'python ..."	About a minute ago	Up About a minute	8000/tcp
	todo-service				
31e31c5c8121	todo-webapp:release-1	"docker-entrypoint.s..."	About a minute ago	Up About a minute	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
000/tcp	webapp				
f306352301b1	todo-notification:release-2	"/bin/sh -c 'python ..."	About a minute ago	Up About a minute	8000/tcp
	notification-service				

4. Web browser

Uber To Do

0 (Tue Aug 25 2026 20:19:14 GMT+0700 (Indochina Time))



 Tue Aug 25 2026 20:19:03 GMT+0700 (Indochina Time)



 Tue Aug 25 2026 20:19:05 GMT+0700 (Indochina Time)

Add



Grader submission

```
(base) PS C:\Users\USER\Desktop\Software_def\act3> .\activity3-windows-CP.exe
2026/08/25 20:14:54 Current Timestamp: 2026-08-25T20:14:54+07:00
2026/08/25 20:14:54 ✓ / Network exists.
2026/08/25 20:14:54 ✓ / Container webapp exists.
2026/08/25 20:14:54 ✓ / Container todo-service exists.
2026/08/25 20:14:54 ✓ / Container notification-service exists.
2026/08/25 20:14:54 ✓ / Container redis exists.
2026/08/25 20:14:54 ✓ / Container api-gateway exists.
2026/08/25 20:14:54 ✓ / All specified containers are running.
2026/08/25 20:14:57 ✓ / Docker compose is running.
2026/08/25 20:15:02 ✓ / URL http://localhost:3000 returns 200.
2026/08/25 20:15:02 ✓ / Todo app is working.
2026/08/25 20:15:02 ✓ / URL http://localhost:8000 returns 404.
2026/08/25 20:15:02 ✓ / URL http://localhost:8000/todo returns 200.
2026/08/25 20:15:02 ✓ / URL http://localhost:8000/notification returns 200.
2026/08/25 20:15:02 ✓ / Notification-service found with api-gateway.
2026/08/25 20:15:02 Result: true
2026/08/25 20:15:02 🎉 Looks good! Please enter your StudentID and Full name below
👉 StudentID: 6630199021
👉 Full Name (TH): เภาสาณ กสิธร
2026/08/25 20:15:20 🎉 Successfully submitted. Your lucky number is 21424113443028378
2026/08/25 20:15:20 🎉🎉🎉 Congratulations! You have completed the activity 🎉🎉🎉
2026/08/25 20:15:20 ⚠️ Don't forget! you still need to submit your assignment via MyCourseVille ⚠️
(base) PS C:\Users\USER\Desktop\Software_def\act3>
```